# COVID-19 Time-Series Data Analysis with pandas

*Cleaning, reshaping, and analysing the Johns Hopkins CSSE global confirmed-cases time series using pandas.*

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## Overview

This project works with the Johns Hopkins CSSE global COVID-19 confirmed-cases dataset, a wide-format time series covering countries and sub-national regions. Using pandas, the raw CSV is loaded, cleaned of missing values, filtered to China's provincial data, reshaped via transposition, and summarised with descriptive statistics.

## Goals

Demonstrate end-to-end data wrangling and exploratory analysis of a real-world epidemiological time series, producing clean, analysis-ready tables and descriptive summaries of provincial case counts.

## Objectives

- Load the global confirmed-cases CSV and audit it for missing values.
- Replace NaN entries with a sentinel string ('not available') to preserve row count.
- Subset the dataset to China and reset the index to a contiguous range.
- Drop non-case columns (Country/Region, Lat, Long) and transpose so provinces become columns and dates become rows.
- Compute total cases per province over the 70-day window and generate a full describe() statistical summary.

## Approach

The analysis follows a linear pandas pipeline: read CSV, inspect for NaNs, apply fillna, boolean-filter rows, drop columns, transpose, then aggregate and describe. Each transformation is validated by displaying intermediate DataFrame states, making the data-flow transparent and reproducible.

## Outcomes

- Produced a clean, transposed province-by-date DataFrame ready for downstream time-series analysis.
- Identified and handled all NaN entries in the raw dataset without dropping any rows.
- Generated per-province total case counts and a full descriptive statistics table across 70 days of data.
- Established a repeatable pandas workflow applicable to any wide-format epidemiological time series.